

Modelling the Shiller CAPE Ratio, Mean Reversion, and Return Forecasts

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KEY FINDINGS

- We provide an estimate of the Shiller cyclically adjusted price/earnings (CAPE) ratio based on economic variables. Our estimate explains more than 90% of the CAPE ratio's variability in the past six decades.
- The conditions for the Shiller CAPE ratio to be mean reverting are not generally met. Notably, inflation, which is an important factor explaining the level and trend of the CAPE ratio, cannot be assumed to be mean reverting.
- For practitioners, our model provides a fair-value estimate of the stock market at any point in time, allows simulations (e.g., of expected returns under a recession scenario), and yields useful forecasts for horizons of three to four years.

ABSTRACT

The Shiller cyclically adjusted price/earnings (CAPE) ratio has yielded overly pessimistic equity return forecasts for two decades, and there is a growing consensus that the CAPE ratio needs to be adjusted for the state of the economy to obtain accurate forecasts. Expanding the literature, the author provides an economic model that explains more than 90% of the CAPE ratio's variability. A link between the economy and CAPE is useful in investment practice. First, the author's model shows which economic variables are most important in driving the CAPE ratio and thus equity returns. Second, a fair-value estimate of the CAPE ratio allows a continuous assessment of whether the stock market is in line with economic fundamentals. Third, the model generates equity return forecasts that are consistent with a practitioner's economic scenario or any alternative scenario such as economic recession. Finally, the author demonstrates that the CAPE ratio is not mean reverting but shows nevertheless how to obtain useful forecasts for horizons of three to four years.

TOPICS

[*Accounting and ratio analysis, simulations, statistical methods, volatility measures**](#)

Making productive use of the Shiller cyclically adjusted price/earnings (CAPE) ratio in investment decision-making processes has been a challenge ever since Campbell and Shiller introduced the measure in 1998. The principal obstacle is that in the past two decades CAPE has almost always been higher than its long-term average. As a result, return forecasts have turned out to be overly

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pessimistic because reversion of the CAPE ratio to its historic mean—a prerequisite for forecasts to produce accurate results—has been elusive.

A number of proposals have been put forward to explain and deal with this conundrum. When it comes to understanding changes of the CAPE ratio (and other valuation measures, such as the price/dividend ratio) over time, Lettau and van Nieuwerburgh (2008) argued that the inconsistent and unstable results of models predicting equity returns using valuation ratios can be reconciled if the assumption of a fixed mean of the state of the economy is relaxed. Cochrane (2011) offered a comprehensive overview of theories and empirical results regarding why discount rates (and thus risk premiums and stock market valuations) vary over time. He also considered macroeconomic approaches to gain further insights into stock market valuations to be promising. Another group of authors have more directly focused on improving return forecasts. Among recent studies, those by Davis et al. (2018) and Arnott, Chaves, and Chow (2017) showed that return forecasts improve when the CAPE ratio is adjusted for the prevailing state of the economy.

The state of the economy is a common thread in the recent literature dealing with CAPE-based return forecasts. We believe that a good understanding of how the state of the economy and the CAPE ratio are linked is paramount to our understanding of the nature of the CAPE ratio and return forecasts. The principal goal of this study is to establish a link between the CAPE ratio and the economy.

In the first (and central) part of the study, we demonstrate that a small set of basic economic variables explains more than 90% of the variation of the CAPE ratio in our sample of monthly data stretching over six decades.¹ We find that historically low inflation and low economic volatility are the key factors explaining why over the past two decades the CAPE ratio has nearly always been higher than its long-term average. Conversely (and surprisingly), changes in real interest rates and earnings growth fail to explain the phenomenon. At the end of the first part, we discuss mean reversion in the context of our model. Simply put, mean reversion of the CAPE ratio requires the state of the economy and all the significant variables describing it to be stationary (mean reverting). This condition is not generally true because inflation—an important variable explaining the level of the CAPE ratio—is a nonstationary variable.² Therefore the CAPE ratio cannot be assumed to be mean reverting.

In the second part of the study we demonstrate that our CAPE model generates better return forecasts than alternative methods, with our forecasts built on the assumption that only the stationary variables explaining the CAPE ratio are expected to be mean reverting over the forecast horizon. The optimal forecast horizon in our framework is half the length of a business cycle, or about 3 to 4 years, and thus appreciably shorter than the standard 10-year horizon (proposed by Campbell and Shiller 1998 and applied by, among others, Davis et al. 2018).

IDENTIFYING THE DRIVERS OF THE CAPE RATIO

As the conceptual framework to identify the drivers of the CAPE ratio, we use an earnings discount model in the form of a Gordon growth model. The net present value P of an equity index at time t is

$$P_t = \frac{E_t}{DF_t - g_t} \quad (1)$$

¹ See Exhibit 3 for an illustration of the result.

² Exhibit 6 depicts inflation in our sample period.

E_t refers to corporate earnings, DF_t to the discount rate, and g_t to the future rate of earnings growth. Decomposing DF_t into the risk-free rate r_t and the equity risk premium z_t and replacing E_t with cyclically adjusted real earnings (the denominator of the CAPE ratio³) yields

$$\frac{1}{\text{CAPE}_t} = r_t + z_t - g_t \quad (2)$$

Taking the natural logarithm (log) and applying the rules for logarithms give rise to

$$\log(\text{CAPE}_t) = -(r_t + z_t - g_t) \quad (3)$$

Equation 3 states that the CAPE ratio increases when the (real) risk-free rate and the equity risk premium fall and (real) earnings growth rises.

DEFINITIONS OF VARIABLES AND DATA SOURCES

Siegel (2016) pointed out that corporate earnings are influenced by changes in accounting rules and that reported earnings in the recent past have been underestimated (and the CAPE ratio overestimated). Philips and Ural (2016) provided an overview of different earnings concepts to calculate the CAPE ratio. They reported that none of the modifications substantially improves CAPE-based equity return forecasts for the US stock market. We thus stick with Shiller's (2017) real earnings data and calculate the CAPE ratio by dividing the (inflation-corrected) month-end value of the S&P 500 (courtesy of Bloomberg) by 10-year average real earnings, lagged three months (our variable is *logCAPE*, the natural logarithm of the CAPE ratio).

As our proxy for the real risk-free rate r we use the yield of 10-year US Treasury notes net of inflation expectations (our variable is *RealBondYield*). We model the latter as the average of the latest available inflation rate and its five-year moving average (our inflation measure is defined later). Nominal yield data are provided by Bloomberg and the Federal Reserve (for data before 1962).

As proxies for expected earnings growth g , we calculate the 5- and 10-year moving averages of gross domestic product (GDP) growth and earnings growth, respectively. Quarterly GDP data are available from the US Bureau of Economic Analysis starting in 1947. We take Shiller's (2017) data to calculate earnings growth rates. We do not find that the 5-year moving averages of GDP growth and earnings growth improve the results, so we present estimates using 10-year moving averages only. (In Appendix A, we describe how we transform quarterly data into monthly data.)

The literature gives a number of suggestions for empirical proxies for the equity risk premium z in Equation 3. The inflation rate is shown to be an important driver of equity valuations (see, e.g., Asness 2003). Our variable *Inflation(log)* is the logarithm of the 12-month percentage change in the US Bureau of Economic Analysis' core consumer expenditure deflator (for further details, see Appendix A), with the core personal consumption expenditures (PCE) deflator also the Fed's preferred inflation measure (see, e.g., Hakkio 2008). Lettau, Ludvigson, and Wachter (2008) found that a persistent decline in the volatility of aggregate consumption growth led to a decline in the equity premium to unprecedented levels in the 1990s. Their findings make the volatility of the economy—we use the broad measure GDP, which also includes, next to consumption, the more volatile corporate investment spending—another

³The CAPE ratio is the ratio of the real (inflation-corrected) value of the S&P 500 and the moving average over the preceding 10 years of real S&P 500 earnings (Shiller 2015).

candidate proxy for the equity risk premium z . Our variables $GDPgrowthVol(5yr,log)$ and $GDPgrowthVol(10yr,log)$ are the 5- and 10-year moving standard deviations of the growth rate in nominal GDP.

That the business cycle (our last proxy for the risk premium) drives stock market valuations is well documented, both theoretically and empirically (see Lucas 1978, Grossman and Shiller 1981, and Campbell and Cochrane 1999). For an overview of variables representing the business cycle, see Darolles, Eychenne, and Martinetti (2010) and Cochrane (2011). We use the unemployment rate as a measure reflecting the economy's stage in the business cycle.⁴ There are a number of arguments in favor of using the unemployment rate (as opposed to, e.g., a consumption-based measure). The unemployment rate represents the entire economy, is an element in most economic forecasts, is widely followed in investment practice, and is a timely available monthly data point. Our variable $UnemploymentRate(log)$ is the natural logarithm of the unemployment rate U-3, published by the US Bureau of Labor Statistics on a seasonally adjusted basis.

A number of studies suggest the use of financial market volatility and other market measures (see Welch and Goyal 2008 for an overview) to explain equity valuations (e.g., Asness 2000 used return volatility to estimate the equity risk premium). Because we focus on the economic drivers of the CAPE ratio, we refrain from using market-based risk measures (we suspect these to be driven by the same economic variables that drive equity valuations). We nevertheless performed some tests. We find that corporate bond spreads do not add to overall results, even though they are an important explanatory variable during the financial crisis of 2008. Adding historic return volatility of the S&P 500 does not improve the overall fit, and the coefficient is not statistically significantly different from zero. Additional details on data, sources, and our use of real-time data are provided in Appendix A.

SAMPLE PROPERTIES

Exhibit 1 depicts selected properties of the variables. The correlation coefficients in Panel A are not high in general, with some exceptions (which we discuss in the next section). All the dependent variables are highly persistent. The coefficients of a first-order autoregressive model (AR[1]), displayed in Panel B, are greater than 0.97 (though none is higher than 1). Persistence should not affect standard errors in our non-small sample, however (Campbell and Yogo 2006; see Appendix B for a further discussion).

The sample consists of stationary and nonstationary variables. The probability that the logarithm of the CAPE ratio has a unit root is 68.7% (Exhibit 1). The augmented Dickey–Fuller unit root test for the inflation rate yields a similar probability. The null hypothesis that the volatility of GDP growth (both the 5- and 10-year measures) has a unit root cannot be rejected either. Conversely, the unemployment rate can be assumed to be stationary. The same is true for the variables $RealBondYield$ and $EarningsGrowth(10yr-MAV)$. An estimate of Equation 3 using the variables in Exhibit 1 requires that the nonstationary variables be cointegrated. The intuition for cointegration is provided in Equation 3 because the explanatory variables of the CAPE ratio (the right-hand side of Equation 3) should cointegrate with the left-hand side, the CAPE ratio. Engle–Granger test results do show that the null hypothesis of no

⁴ We also tested de-trended US leading indicators (the Index of 10 Leading Indicators, published by the US Conference Board) as an indicator for the business cycle, but we found that results were slightly worse compared with the estimates shown in Exhibit 2.

EXHIBIT 1

Variable Characteristics

	logCAPE	RealBondYield	Inflation(log)	Inflation(log,5yr-MAV)	GDPgrowthVol(5yr,log)	GDPgrowthVol(10yr,log)	UnemploymentRate(log)	GDPgrowth(10yr-MAV)	EarningsGrowth(10yr-MAV)
Panel A: Correlations									
logCAPE	1.00								
RealBondYield	-0.20	1.00							
Inflation(log)	-0.82	0.23	1.00						
Inflation (log,5yr-MAV)	-0.83	0.43	0.85	1.00					
GDPgrowthVol (5yr,log)	-0.73	0.01	0.40	0.41	1.00				
GDPgrowthVol (10yr,log)	-0.65	0.06	0.35	0.31	0.80	1.00			
UnemploymentRate(log)	-0.53	-0.02	0.22	0.45	0.49	0.24	1.00		
GDPgrowth (10yr-MAV)	-0.11	0.26	0.30	0.11	0.18	0.21	-0.51	1.00	
EarningsGrowth (10yr-MAV)	0.36	-0.20	-0.13	-0.37	-0.24	-0.26	-0.44	0.13	1.00
Panel B: Variable Characteristics									
Mean	2.92	2.85	1.59	1.59	1.82	1.94	1.77	3.24	1.87
Minimum	1.90	-0.29	1.08	1.09	1.03	1.25	1.22	1.32	-15.39
Maximum	3.83	7.92	2.50	2.27	2.57	2.70	2.38	5.11	9.62
Standard deviation	0.40	1.53	0.36	0.34	0.40	0.36	0.25	0.82	3.24
Autocorrelation (AR[1])	0.995	0.978	0.997	0.999	0.998	0.997	0.992	1.000	0.993
Probability of a unit root (%)	68.7	2.7*	69.9	34.6	37.1	28.0	1.1*	39.8	0.1**

NOTES: The sample includes monthly data, June 1957 to June 2017 (for the variables using GDP, the latest quarterly data point available in any given month is taken, as described in the section Definitions of Variables and Data Sources). Panel A shows correlation coefficients. Panel B shows standard statistics. Augmented Dickey-Fuller tests are applied to test for unit roots (* and ** denote rejection of a unit root at the 95% and 99% confidence levels, respectively).

cointegration of the nonstationary variables can be rejected at the 95% confidence level for all the specifications of Equation 3 shown in Exhibit 2.

ESTIMATION RESULTS

The empirical form of Equation 3 is

$$\log\text{CAPE}_t = \alpha + \sum_{i=1}^k \beta_i \cdot x_{i,t} + \varepsilon_t \quad (4)$$

with α a constant, β_i the coefficients of the k independent variables x_i ($i = 1, \dots, k$), and ε an independently and identically distributed (i.i.d.) error term. We estimate the parameters of Equation 4 by ordinary least squares (OLS). Least squares estimates are consistent (even super-consistent; see Stock 1987) in our non-small sample of 721 monthly observations over six decades. Standard errors are Newey–West corrected, given serial correlation in the residuals. We provide a discussion of alternative estimation methods in Appendix B.

EXHIBIT 2

Regression Results

Dependent Variable: logCAPE						
Estimate	1	1a	2	2a	3	3a
Standardized Variables?	No	Yes	No	Yes	No	Yes
Independent variable	Coefficient (standard error)	Coefficient (standard error)	Coefficient (standard error)	Coefficient (standard error)	Coefficient (standard error)	Coefficient (standard error)
Intercept	5.15** (0.14)		5.70** (0.11)		5.68** (0.09)	
RealBondYield	0.00 (0.01)	0.01 (0.03)	−0.01 (0.01)	−0.04 (0.03)		
Inflation(log)	−0.50** (0.06)	−0.44** (0.05)	−0.69** (0.04)	−0.61** (0.03)	−0.70** (0.04)	−0.62** (0.03)
Inflation (log,5yr–MAV)	−0.28** (0.07)	−0.24** (0.06)				
GDPgrowthVol (5yr,log)	−0.26** (0.04)	−0.26** (0.04)				
GDPgrowthVol (10yr,log)	−0.21** (0.04)	−0.19** (0.04)	−0.39** (0.03)	−0.35** (0.03)	−0.39** (0.03)	−0.35** (0.03)
UnemploymentRate(log)	−0.14* (0.06)	−0.09* (0.04)	−0.51** (0.06)	−0.32** (0.04)	−0.50** (0.05)	−0.31** (0.03)
GDPgrowth (10yr–MAV)	0.04** (0.02)	0.08** (0.03)	0.00 (0.02)	0.00 (0.03)		
EarningsGrowth (10yr–MAV)	0.01 (0.00)	0.05 (0.03)				
Adjusted R ²	0.93		0.90		0.90	
Regression standard error	0.11		0.12		0.13	
Durbin–Watson statistic	0.18		0.16		0.16	
No. of observations	721		721		721	

NOTES: Parameters are estimated by ordinary least squares, with standard errors Newey–West corrected. The sample period is June 1957 to June 2017. The coefficient is different from zero at the *95% and **99% confidence level. Data sources are described in the text and the data appendix.

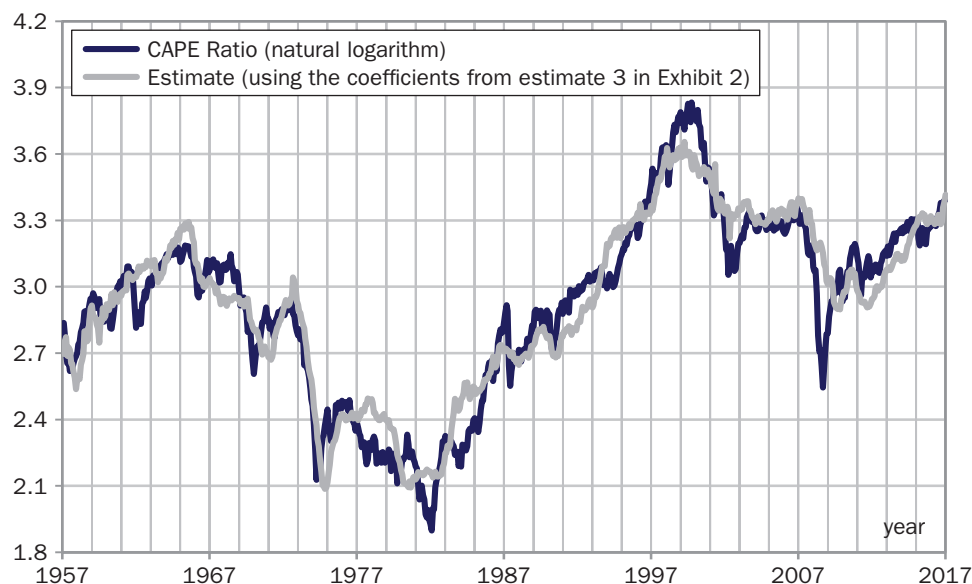
Estimate 1 in Exhibit 2 shows the results of a regression that includes all the variables from Exhibit 1. To assess the relative importance of the variables, we also provide results using standardized variables (these estimate numbers carry the suffix a). The adjusted R² of 0.93 shows a good (and better-than-expected) fit. Looking at the individual variables, the *RealBondYield* exerts surprisingly little influence on the CAPE ratio, which contrasts with a number of previous studies (see, e.g., Asness 2003). The *RealBondYield*'s coefficient is far from statistical and economic significance (the standardized coefficient is a mere 0.01). Among the growth proxies, *GDPgrowth(10yr–MAV)* has the expected positive sign and is statistically different from zero (at the 99% confidence level), whereas corporate earnings growth (*EarningsGrowth[10yr–MAV]*) is not significant.

The variables representing inflation and the volatility of GDP growth have the highest standardized coefficients, with all the coefficients having the expected negative sign. The business-cycle variable *UnemploymentRate(log)* has the expected negative sign and is statistically significant at the 95% level; however, its standardized coefficient is considerably smaller than those of inflation and economic volatility.

One of our objectives is to isolate the influence of the business cycle as clearly as possible. Exhibit 1 shows that *UnemploymentRate(log)* has comparatively high

EXHIBIT 3

CAPE Ratio and Estimate

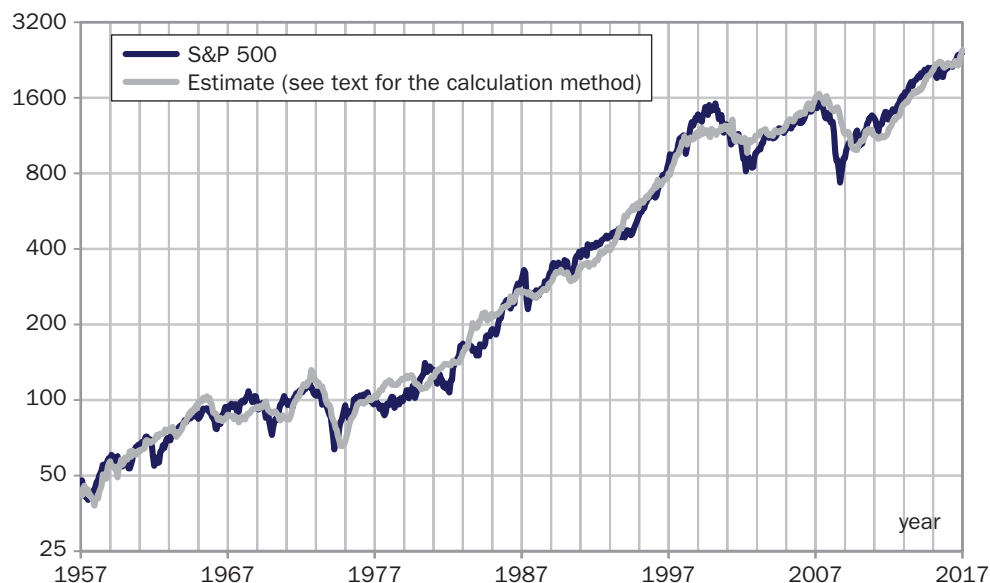
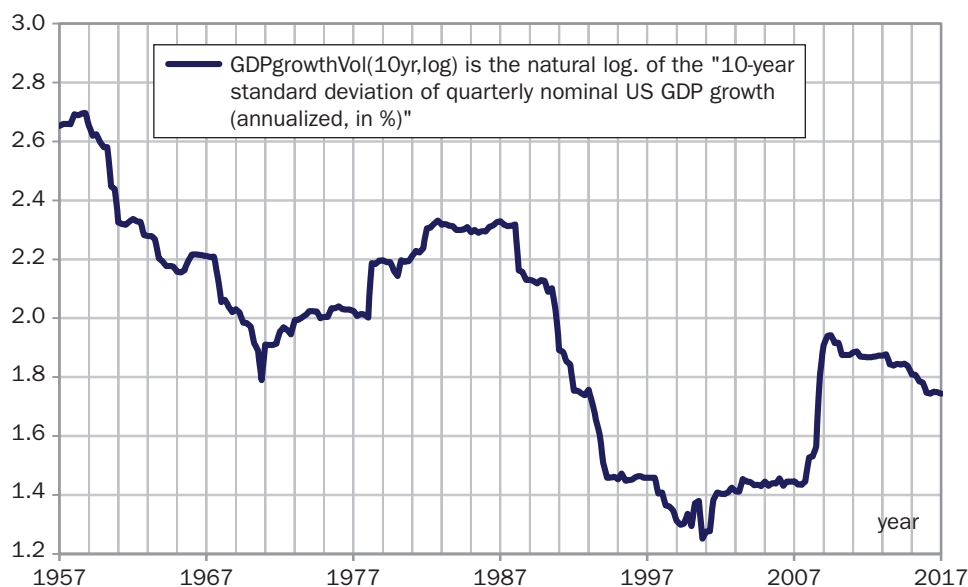


correlations with the variables *Inflation(log,5yr-MAV)*, *GDPgrowthVol(5yr,log)*, *GDPgrowth(10yr-MAV)*, and *EarningsGrowth(10yr-MAV)*, respectively. Therefore, in estimate 2, we eliminate *Inflation(log,5yr-MAV)* and *GDPgrowthVol(5yr,log)*. We also exclude the insignificant growth proxy *EarningsGrowth(10yr-MAV)*. The adjusted R^2 drops moderately from 0.93 in estimate 1 to 0.90 in estimate 2. After further eliminating the insignificant variables *RealBondYield* and *GDPgrowth(10yr-MAV)* in estimate 2 we arrive at the parsimonious estimate 3, which we use in the remainder of the study. In Appendix C, we provide evidence that regression parameters are stable over time.

Exhibit 3 illustrates how estimate 3 tracks the CAPE ratio over time. The estimate captures the rise of the CAPE ratio early in the sample (through the mid-1960s), its subsequent fall to the sample low in 1982, the long stretch of ever rising valuations through 2000, and much of the variation in the CAPE ratio thereafter. Transforming the CAPE estimate into a fair-value estimate of the S&P 500 (see Exhibit 4) is achieved by multiplying the CAPE estimate and CAPE earnings.

The results (estimate 3, Exhibit 2) can be summarized as follows. Given an adjusted R^2 of 0.90, the estimate explains much of the variability of the CAPE ratio. That estimate 3 is quite a good model of the CAPE ratio throughout the sample is illustrated in Exhibit 3 (the same conclusion holds for its transformation into a model of the S&P 500, as visualized in Exhibit 4). The coefficients for *Inflation(log)*, *GDPgrowthVol(10yr,log)*, and the *UnemploymentRate(log)* have the expected negative sign and are significant at the 99% level. Given a high standardized coefficient (and incremental R^2 values pointing in the same direction, not shown), there is good evidence that inflation is the most important variable driving the CAPE ratio over time.

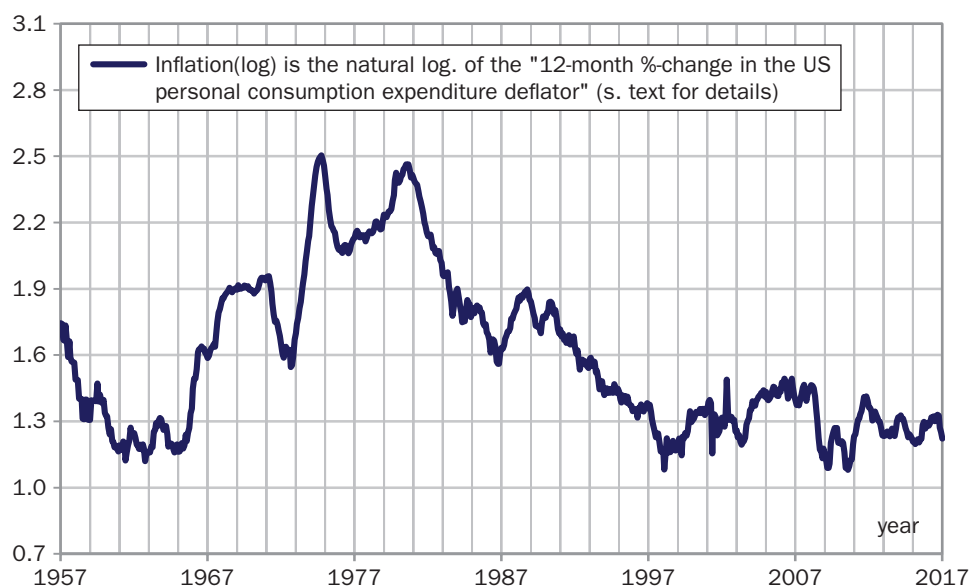
Our model is consistent with the CAPE ratio almost always being higher than its long-term average in the past two decades. A more stable economy than in the past, characterized by a secular downtrend in macroeconomic volatility (see Exhibit 5)—often dubbed the Great Moderation (Summers 2005)—and a low inflation rate (Exhibit 6) imply a historically low level of perceived risks and a high CAPE ratio. Conversely, earnings growth and real interest rates fail to explain the rise in the CAPE ratio in recent decades.

EXHIBIT 4**S&P 500 and Estimate (log. scale)****EXHIBIT 5****Variable GDPgrowthVol(10yr,log)**

Our results suggest that the CAPE ratio can be considered to be an endogenous variable in an economic model. Put differently, the CAPE ratio reflects how investors perceive the economy, notably economic risks. Our model also says that any macroeconomic forecast (to the extent that it includes a forecast of inflation, the unemployment rate, and GDP) implies a fair-value estimate of the CAPE ratio and the stock market. This feature of the model is useful in investment practice to the extent that it helps to align a practitioner's economic forecast and his or her equity market forecast.

EXHIBIT 6

Variable Inflation(log)



Multivariate estimates of the CAPE ratio's level are not readily available in the literature, so we compare our results with a fair-value estimate of the CAPE ratio proposed in a recent study by Davis et al. (2018), acknowledging that the authors' primary goal is to improve equity return forecasts and not to explain the level of the CAPE ratio. Using their variables (i.e., the real bond yield, inflation; "realized S&P 500 price volatility, over trailing 12 months"; and the 12-month volatility of changes in real bond yields), we get an adjusted R^2 in our sample period of 0.67 (with not all variables having the expected sign⁵). When we add the variables $GDPgrowthVol(10yr,log)$ and $UnemploymentRate(log)$, the adjusted R^2 increases to 0.88.⁶ This improvement in the R^2 is significant, not only statistically but importantly also to our attempt to understand the nature and key driving forces of the CAPE ratio.

CONDITIONS FOR MEAN REVERSION OF THE CAPE RATIO

The condition for mean reversion of the CAPE ratio is given in Equation 3. The CAPE ratio is stationary (and mean reverting) when the right-hand side of Equation 3 is stationary. Consequently, the independent variables, at least those exerting significant influence on the CAPE ratio, all need to be stationary. If one or more of the drivers of the CAPE ratio are nonstationary, the CAPE ratio is nonstationary. In our sample, the CAPE ratio is nonstationary because two of its significant drivers, inflation and macroeconomic volatility, are nonstationary. This finding gives rise to the proposition that the CAPE ratio cannot be assumed to be stationary a priori.

⁵The variable "realized S&P 500 price volatility, over trailing 12 months" (Davis et al. 2018) is significant in our estimate (at the 99% confidence level) but does not have the expected positive sign (regressed on the logarithm of 100/CAPE as done by Davis et al. 2018).

⁶The difference from the R^2 of 0.90 in estimate 3 (Exhibit 2) stems from a different inflation measure. In Exhibit 2, we use core PCE inflation, whereas the regression to compare the two estimates takes the Consumer Price Index, as done by Davis et al. (2018).

A NEW FRAMEWORK TO FORECAST RETURNS

To obtain a forecast of the CAPE ratio (and thus the stock market), forecasts of inflation, the unemployment rate, and macro-volatility are required. For the purpose of back testing, we use the expected future values of the explanatory variables consistent with their data-generating processes. The expected future value of a stationary variable is its mean.⁷ The only stationary variable (in our most parsimonious estimate) is the unemployment rate, and we take the variable's 30-year moving median to estimate its mean. As for the nonstationary variables, inflation and macroeconomic volatility, we assume that they follow a random walk ($y_{t+1} = y_t + u_t$, with u_t being an i.i.d. error term with expected value $E[u_t] = 0$). Hence, y_t is the expected future value $E(y_{t+h})$ at time t (for $h > 0$).⁸

With forecasts for the independent variables at hand, we proceed to calculate the *cycle-adjusted* (i.e., adjusted for business cycle fluctuations) fair value of the CAPE ratio, $E(\text{CAPEca})$. To avoid look-ahead bias, the coefficients (estimate 3, Exhibit 2) are re-estimated recursively each month using real-time economic data. The first recursive estimate is for the period June 1957 through June 1977. Out-of-sample forecasts start in July 1977 and end in June 2017 (net of the forecast horizon).

The forecast of the real annualized price returns of the S&P 500, $E(r_{t+h})$ in Equation 5, consists of real annual earnings growth, eg_{t+h} (estimated by its historic⁹ mean at time t), minus the fraction C ($0 \leq C \leq 1$) of the term in square brackets, the difference of $\log \text{CAPE}_t$ and its *cycle-adjusted* fair value $E(\log \text{CAPEca}_t)$.

$$E(r_{t+h}) = eg_{t+h} - C \cdot [\log \text{CAPE}_t - E(\log \text{CAPEca}_t)] \quad (5)$$

We calculate C as half the average length of the business cycle. In our sample, the average time from peak to peak in the unemployment rate is 83 months. The CAPE ratio thus can be expected to revert to its cycle-adjusted fair value in 42 months (half of 83 months, rounded), on average. Therefore, forecasts that incorporate business cycle information should do comparatively well over horizons half the length of a typical business cycle, which is approximately three to four years (this conclusion aligns with the findings of Lettau and Ludvigson 2001). Hence, the convergence parameter C is the forecast horizon h (in months) divided by 42 months (C is 1 for $h > 42$).

We compare our cycle-adjusted CAPE-based equity return forecast with three alternative methods. The first is a classic CAPE return forecast in the form

$$r_{t+h} = \alpha_{h,t} + \beta_{h,t} \cdot \log \text{CAPE}_t + \varepsilon_{h,t} \quad (6)$$

with ε an i.i.d. error term. The parameters $\alpha_{h,t}$ and $\beta_{h,t}$ are estimated by OLS recursively at t for each forecast horizon h .

The second set of alternative forecasts is based on Lettau and Ludvigson's (2001) consumption to wealth ratio, CAY , in conjunction with the dividend-to-price ratio (DP):

$$r_{t+h} = \alpha_{h,t} + \beta_{h,t} \cdot \text{CAY}_t + \gamma_{h,t} \cdot \text{DP}_t + \varepsilon_{h,t} \quad (7)$$

We use CAY data from Lettau (2018) and estimate the parameters $\alpha_{h,t}$, $\beta_{h,t}$, and $\gamma_{h,t}$ recursively (by OLS). Data for the dividend-to-price ratio DP are obtained from Shiller (2017).

⁷ This applies whenever the forecast horizon is long enough for the variable to return to its mean.

⁸ It is worth noting that y_t is not generally a good forecast of y_{t+h} , given the nature of the random walk process.

⁹ The (somewhat arbitrary) start date is 1925. The same start date is used for the recursive estimates of Equation 6 and to obtain average historic returns for the naïve forecast.

EXHIBIT 7

Forecast Evaluation

	Cycle-Adjusted CAPE		Classic CAPE			CAY/Dividend- Price Ratio			Naïve		
	ME	MAE	ME	MAE	Prob.	ME	MAE	Prob.	ME	MAE	Prob.
Forecast Horizon (years)											
2	-1.40	8.27	-4.66	11.27	0.0**	-3.77	9.71	0.0**	-2.14	8.86	2.0*
3	-1.65	6.82	-4.67	9.20	0.0**	-3.40	7.60	0.3**	-2.29	7.83	0.0**
4	-1.93	6.10	-4.57	7.71	0.0**	-3.40	6.38	21.5	-2.31	7.24	0.0**
5	-2.29	5.35	-4.49	6.22	0.0**	-3.91	5.49	42.0	-2.39	6.42	0.0**

NOTES: Forecast period 1977 to 2017 (net of forecast horizon). The forecasts are explained in the text. ME is the mean annual forecast error and MAE is the mean annual absolute forecast error. *Prob.* refers to the Diebold–Mariano test probability (in percent) that the MAE of the cycle-adjusted CAPE forecast is the same as the forecast evaluated. * (**) denotes a probability lower than 5% (1%).

Finally, we compare our forecasts with a naïve forecast, which is simply the recursive mean of the price return at time t .

FORECAST RESULTS

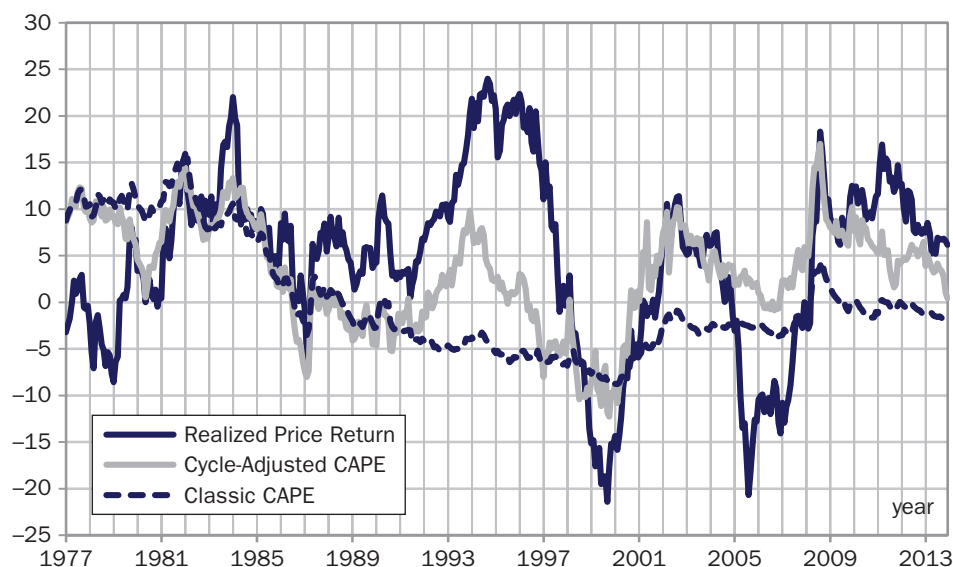
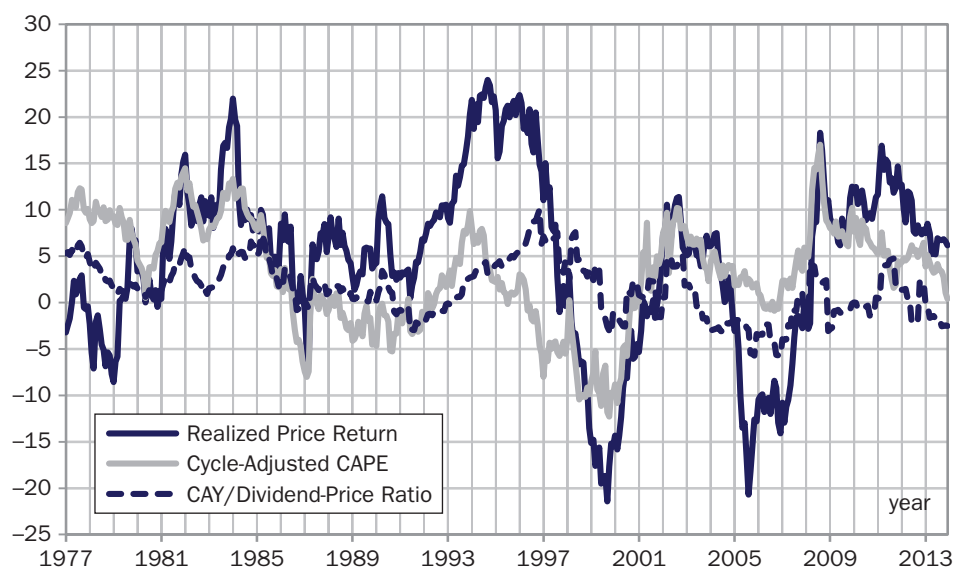
The results of the four forecast methods for two- to five-year returns are summarized in Exhibit 7. For each forecast, the exhibit contains the mean forecast error (ME) and the mean absolute error (MAE). For each of the three alternative forecasts (classic CAPE, CAY/dividend–price ratio, and naïve), we provide probabilities of the MAEs being the same as those of the cycle-adjusted CAPE forecast based on the Diebold–Mariano test.

The ME of the cycle-adjusted CAPE ratio forecast is the lowest (in absolute terms) of the four forecasts methods for all forecast horizons. The second-best MEs we get from the naïve forecast. The CAY/dividend–price ratio method comes in third, followed by classic CAPE. The MAEs of the cycle-adjusted CAPE forecasts are below the MAEs of the other forecasts for all forecast horizons. The difference in the MAEs is statistically significant at the 99% confidence level when comparing the cycle-adjusted CAPE with the classic CAPE forecast. This result only confirms previous studies that demonstrate that forecasts improve when the CAPE ratio is adjusted for the state of the economy (e.g., Lettau and van Nieuwerburgh 2008; Arnott, Chaves, and Chow 2017; and Davis et al. 2018).

Similarly, the MAEs are statistically significantly lower than the naïve forecasts (at the 95% confidence level for two-year forecasts and at the 99% level for three-, four- and five-year forecasts). Although the MAEs of the cycle-adjusted CAPE forecast are lower than those of the CAY/dividend–price ratio forecast, the difference is significant for the two- and three-year forecasts but not the four- and five-year horizons.

How the forecast patterns differ for a three-year forecast horizon is illustrated in Exhibits 8 and 9. The exhibits show the cycle-adjusted CAPE forecast and the realized three-year forward price return. Exhibit 8 also contains the classic CAPE forecast and Exhibit 9 the CAY/dividend–price ratio forecast, respectively.

The cycle-adjusted model captures the increase in returns during the 1990s, for instance, and forecasts negative returns at the peak of the technology bubble in the year 2000 (with the forecast even modestly more accurate—that is, more negative—than the classic CAPE model, despite the latter underestimating returns most, on

EXHIBIT 8**S&P 500, Three-Year Forward Returns (% p.a.)****EXHIBIT 9****S&P 500, Three-Year Forward Returns (% p.a.)**

average, in the period 1977 to 2017). Conversely, because there is no marked overvaluation of the stock market before the financial crisis of 2008, the cycle-adjusted CAPE model only forecasts below-average returns (because the unemployment rate is lower than its long-term average in 2007) but not negative returns. Exhibit 9 reveals that the forecast patterns of the cycle-adjusted CAPE and the CAY/dividend-price ratio models at times differ markedly—for example, between 1995 and 2001. (An analysis of why the forecast patterns differ is beyond the scope of this study, however.)

ECONOMIC SCENARIOS AND THE FAIR VALUE OF THE CAPE RATIO

Our model (using the parameters of estimate 3 in Exhibit 2) permits calculation of the fair value of the CAPE ratio (and thus the stock market) for any economic scenario. An example of a scenario is a cycle-adjusted fair value of the CAPE ratio using the latest value of *GDPgrowthVol(10yr,log)* in the sample, the 30-year median of the unemployment rate, and 2% for inflation (the Federal Reserve's inflation target). In this scenario, the fair value of the CAPE ratio is 23.6 (21% lower than the latest value in the sample, but 40% higher than the CAPE ratio's mean of 16.8 in the period 1871–2017). The value of 23.6 is the mean of the CAPE ratio to which an investor would expect the ratio to revert if assuming that inflation will in the future average 2% and that economic volatility will, on average, correspond to the value at the end of our sample (which is also close to the variable's 30-year mean). In a related forecast, we use the same values, except for the unemployment rate, for which we take the value it typically assumes in a recession.¹⁰ The fair value of the CAPE ratio in this scenario is 21.3. This is 29% less than the latest value in the sample, although still appreciably higher than the CAPE ratio's historic mean of 16.8.

ADJUSTED CAPE MODELS: A NOTE OF CAUTION

A number of studies (including the present study) show that equity return forecasts using the CAPE ratio improve when the ratio is adjusted for the state of the economy. Such adjusted models¹¹ provide better forecasts when the drivers of the CAPE ratio assume values markedly different from historic averages for a protracted period—say, a number of decades, possibly after a structural break (for a discussion of structural breaks, see Lettau and van Nieuwerburgh 2008). This has been the case since the Great Moderation: Inflation and economic volatility have been lower since the 1990s than in the past 100 years. If, on the other hand, inflation and economic volatility revert back to their historic mean (or move higher), adjusted CAPE models will underperform a classic CAPE framework that assumes reversion to a long-term historic mean (with such a mean consistent with a higher inflation rate and higher macro-volatility than in the past two decades). Put differently, stock market forecasts using a CAPE ratio adjusted for the state of the economy rest on the assumption that the deviation of the state of the economy from its historic mean will persist. At present, the implicit assumption is that the Great Moderation is a lasting phenomenon.

SUMMARY AND CONCLUSION

We provide a fair-value estimate of the CAPE ratio and find that three variables—inflation, the volatility of economic growth, and the unemployment rate—explain 90% of its variability. This finding allows the level of the CAPE ratio to be interpreted as representing investors' perceptions of the state of the economy. When economic well-being is high (i.e., inflation and the unemployment rate are low) and the economy is perceived to be stable (macroeconomic volatility is low), the CAPE ratio is high (and

¹⁰In our sample, the unemployment rate peaks, on average, at 2.2% above the non-accelerating inflation rate of unemployment (NAIRU) during recessions. Adding 2.2% to NAIRU (which was 4.7% in June 2017, according to the Congressional Budget Office) yields 6.9%.

¹¹Most models that adjust for the state of the economy do so by taking the latest value or lagged values of the variables representing the state of the economy. In our model, we distinguish between stationary and nonstationary variables, with only the latter implying the use of the most recently available value for the variable as its forecast.

vice versa). Our model identifies historically low inflation and low economic volatility as the key factors explaining why over the past two decades the CAPE ratio has nearly always been higher than its long-term average. Conversely, changes in real interest rates and earnings growth fail to explain the phenomenon.

Our analysis shows that the CAPE ratio has a steady mean only when its significant economic drivers are stationary. In our sample, inflation and macroeconomic volatility are nonstationary and, hence, the CAPE ratio is nonstationary. These findings suggest that the CAPE ratio cannot be assumed to be stationary a priori.

To forecast equity returns, our framework requires a forecast of the economic drivers of the CAPE ratio (these are the mean for stationary variables and the latest available value for nonstationary variables). Our out-of-sample forecasts based on recursive estimates and real-time data show that we get superior results compared with alternative methods for horizons that are half the length of a typical business cycle, or about three to four years.

We believe that our model that explains the level of the CAPE ratio and its trends over time has valuable applications in investment practice. First, we show that there is a strong link between the economy and the CAPE ratio. We point to those economic variables that are most important in driving the CAPE ratio and thus equity returns, both in the short- and in the long-run. Second, a fair-value estimate of the CAPE ratio allows a continuous assessment of whether the stock market is in line with economic fundamentals. Third, investors may use our model to get to an equity return estimate that is consistent with their economic scenario. Alternative scenarios, such as an economic recession or enduring expansion, can be simulated as well. Finally, our framework is simple and transparent, and our model relies on timely and readily available economic data, which we believe is important for a tool to be suitable in practice.

APPENDIX A

DATA

Our study is based on a set of monthly data from June 1957 through June 2017. The independent variables are in percentages. We take the natural logarithm (log) of the variables for which the literature or preliminary regressions suggest logarithmic relationships.

We use variables with the lags indicated by their release schedules. The unemployment rate (U-3) and the inflation rate are monthly data released within four weeks after month end. The core PCE index is available beginning in January 1959. For January 1958 to December 1959, we take the US Bureau of Labor Statistics' core Consumer Price Index and before that the Consumer Price Index from the same source. We add the constant 2 (before taking the logarithm of the inflation rate) to avoid allowing a change in inflation to have a disproportionately large influence on the CAPE ratio when the inflation rate is very low.

We obtain monthly GDP data points (to calculate the moving averages of GDP growth and the volatility of GDP growth) by taking the most recent quarterly GDP data point available at any given month end (e.g., for March 2017, we take GDP data for the fourth quarter 2016). We do not interpolate the data.

We use real-time data—that is, the data that were available at the end of any sample period (e.g., for the full sample ending in June 2017, we take the data that were available at the end of June 2017). Vintages of data series are provided by the Federal Reserve banks of Philadelphia and St. Louis.

APPENDIX B

ESTIMATION METHOD

The parameters of Equation 4 can be estimated through several methods. A cointegrating equation of a vector error correction model is not pursued because autocorrelation in the underlying VAR model persists, even when the lag length is long (up to 36 months). Parameter estimates also turn out to be sensitive to the lag length used (for this phenomenon, see Stock and Watson 1993).

When we compare the results from OLS and dynamic least squares (Stock and Watson 1993), all with Newey–West-corrected standard errors, we find that the results (coefficients and, notably, standard errors) are strikingly similar. Nevertheless, the coefficients estimated by dynamic least squares are somewhat sensitive to the lag and lead length chosen (different from what Lettau and Ludvigson 2001 reported).

Least squares estimates are consistent (even super-consistent, see Stock 1987) in our non-small sample of 721 monthly observations over six decades. The standard errors obtained from OLS estimates are biased, given serial correlation in the residuals (Newey–West standard errors eliminate the bias). In addition, the residuals obtained from a first-order autoregressive (AR[1]) estimate of *GDPgrowthVol* (both the 5-year and 10-year rolling standard deviations) are correlated with the error term in Equation 4 (the correlation coefficients are approximately -0.3). However, this should not affect standard errors in our non-small sample (Campbell and Yogo 2006).

Because we are primarily interested in a consistent set of coefficients, we estimate Equation 4 by OLS. To assess possible bias in the Newey–West-corrected standard errors, we perform Monte Carlo simulations. In the simulations, we apply data-generating processes that mirror the variables in the sample (highly persistent stationary and nonstationary variables, with one nonstationary variable's innovations correlated with the error term in Equation 4). A high serial correlation in the residuals (low Durbin–Watson statistic; see Exhibit 2) suggests the presence of omitted variables, which we model as highly persistent but not correlated with the regressors. The simulations show that the coefficients are unbiased, as expected. The Newey–West-corrected standard errors appear to underestimate the standard errors of the nonstationary variables when nonstationary omitted variables are present (nonstationary omitted variables are not sufficiently important in the simulation to jeopardize the assumption of stationary residuals, however). We find that the simulated standard errors of the coefficients are about twice as high in a sample with 721 observations and in a sample half that size, irrespective of whether the innovations of the nonstationary variables correlate with the error term in Equation 4. We note, however, that the conclusions drawn in this study do not change if we assume that the standard errors are twice the size of those shown in Exhibit 2.

APPENDIX C

ESTIMATES FOR TWO NON-OVERLAPPING SUBSAMPLES

Exhibit C1 shows the specification of estimate 3 in Exhibit 2 for two non-overlapping subsamples, June 1957 through June 1987 and July 1987 through June 2017 (the full sample, 1957 through 2017, shows the values from Exhibit 2).

The coefficients for *Inflation(log)*, *GDPgrowthVol(10yr,log)*, and *UnemploymentRate(log)* are statistically significantly different from zero at the 99% level in both subsamples (the standard errors are Newey–West corrected). When testing for differences, we find that the coefficients are not statistically different in the two subsamples, except for *GDPgrowthVol(10yr,log)*, which is significantly lower in the second subsample than in the full sample (at the 95% confidence level, without accounting for possible bias as suggested in Monte Carlo simulations; see the main text and Appendix B).

EXHIBIT C1

Regression Results for Subsamples

Dependent Variable: CAPE(log)			
Sample	1957–1987 Coefficient (standard error)	1987–2017 Coefficient (standard error)	1957–2017 Coefficient (standard error)
Intercept	5.56** (0.19)	5.56** (0.16)	5.68** (0.09)
Inflation(log)	−0.67** (0.04)	−0.70** (0.09)	−0.70** (0.04)
GDPgrowthVol (10yr,log)	−0.34** (0.07)	−0.25** (0.06)	−0.39** (0.03)
UnemploymentRate(log)	−0.52** (0.06)	−0.56** (0.08)	−0.50** (0.05)
Adjusted R ²	0.88	0.79	0.90
Regression standard error	0.12	0.12	0.13
Durbin–Watson statistic	0.17	0.17	0.16
No. of observations	361	360	721

NOTE: Parameters are estimated by ordinary least squares, with standard errors Newey–West corrected. **The coefficient is different from zero at the 99% confidence level. The data and sources are described in the text and the data appendix.

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